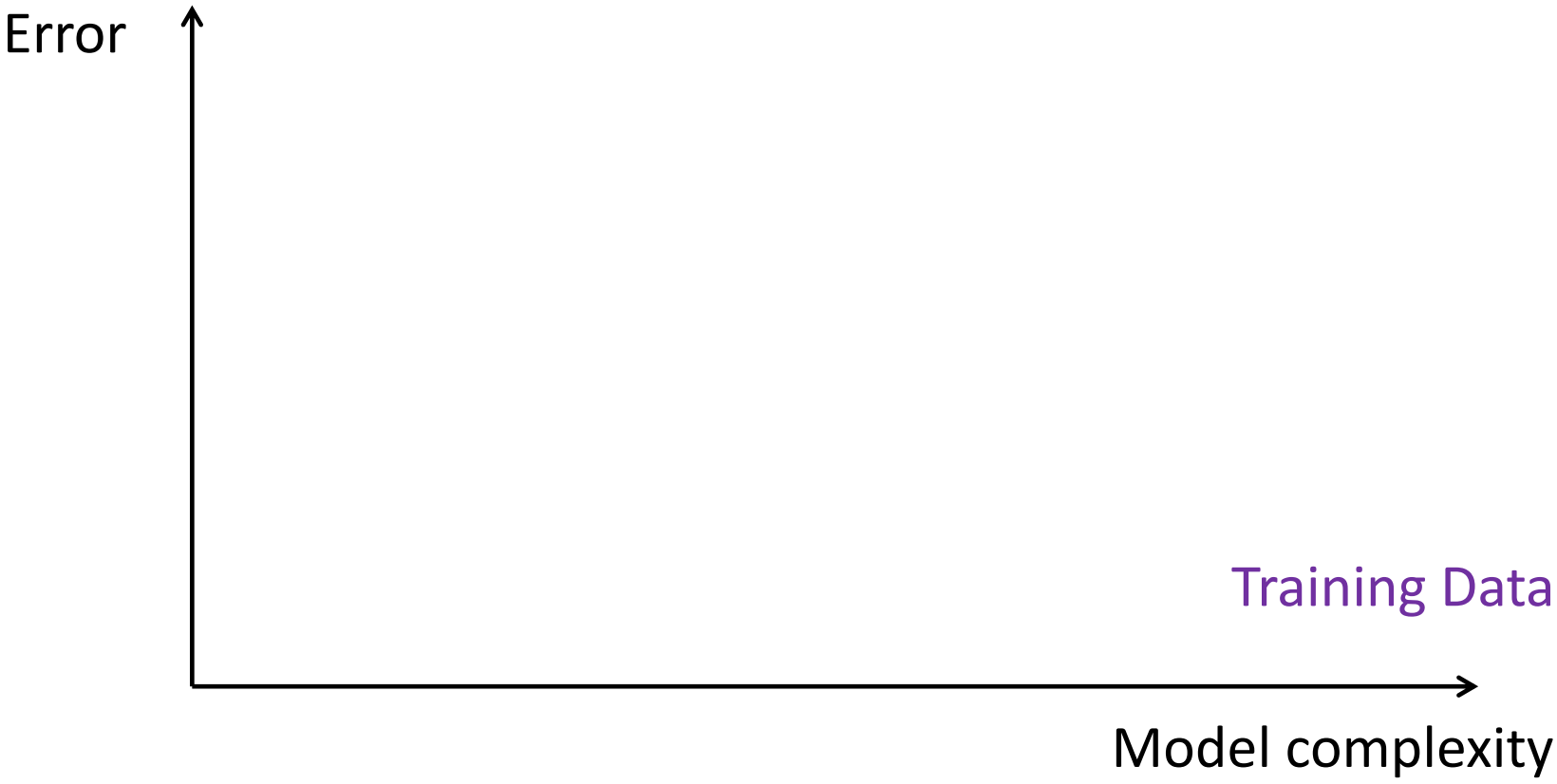
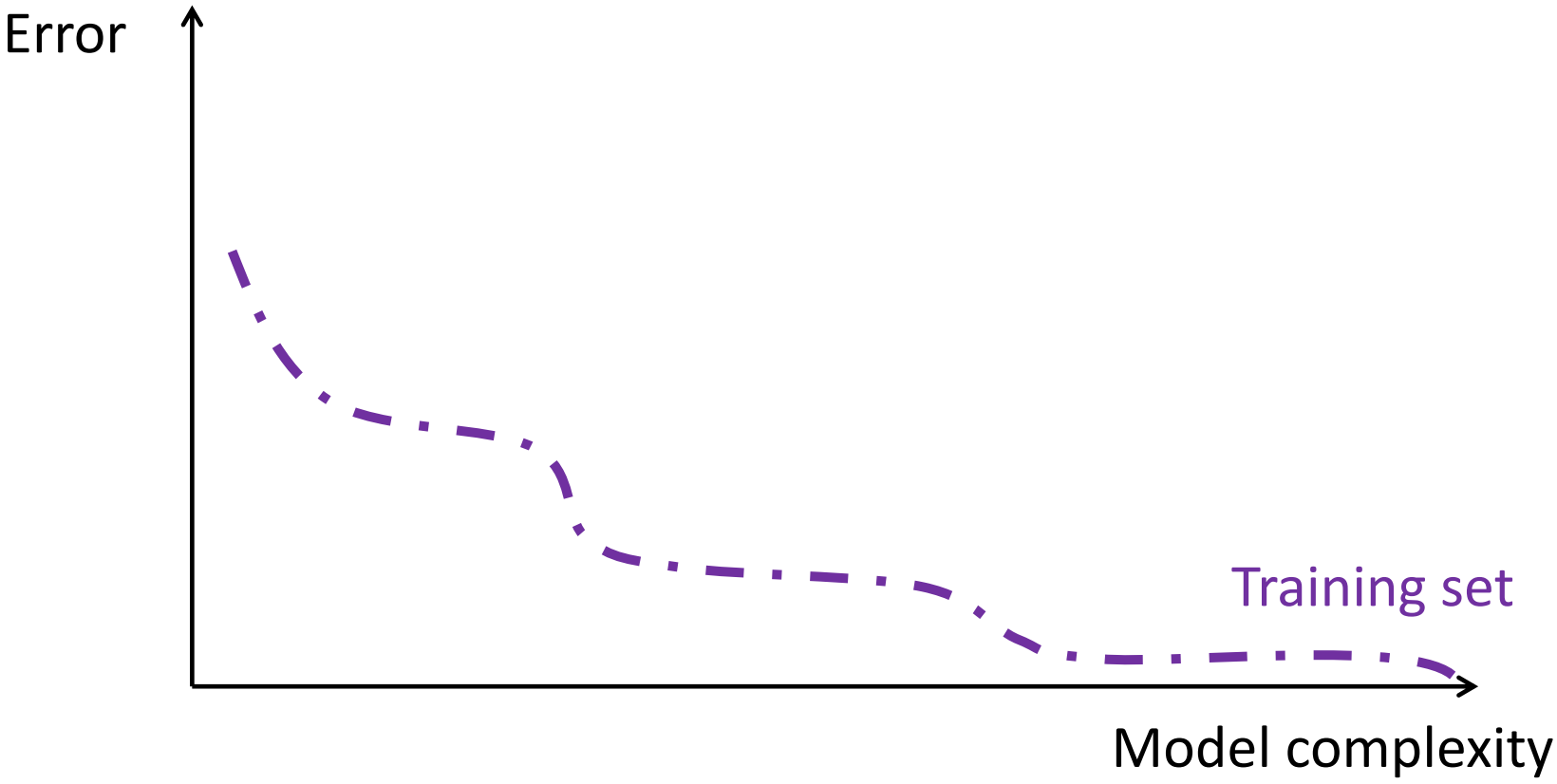
Overfitting and Regularization

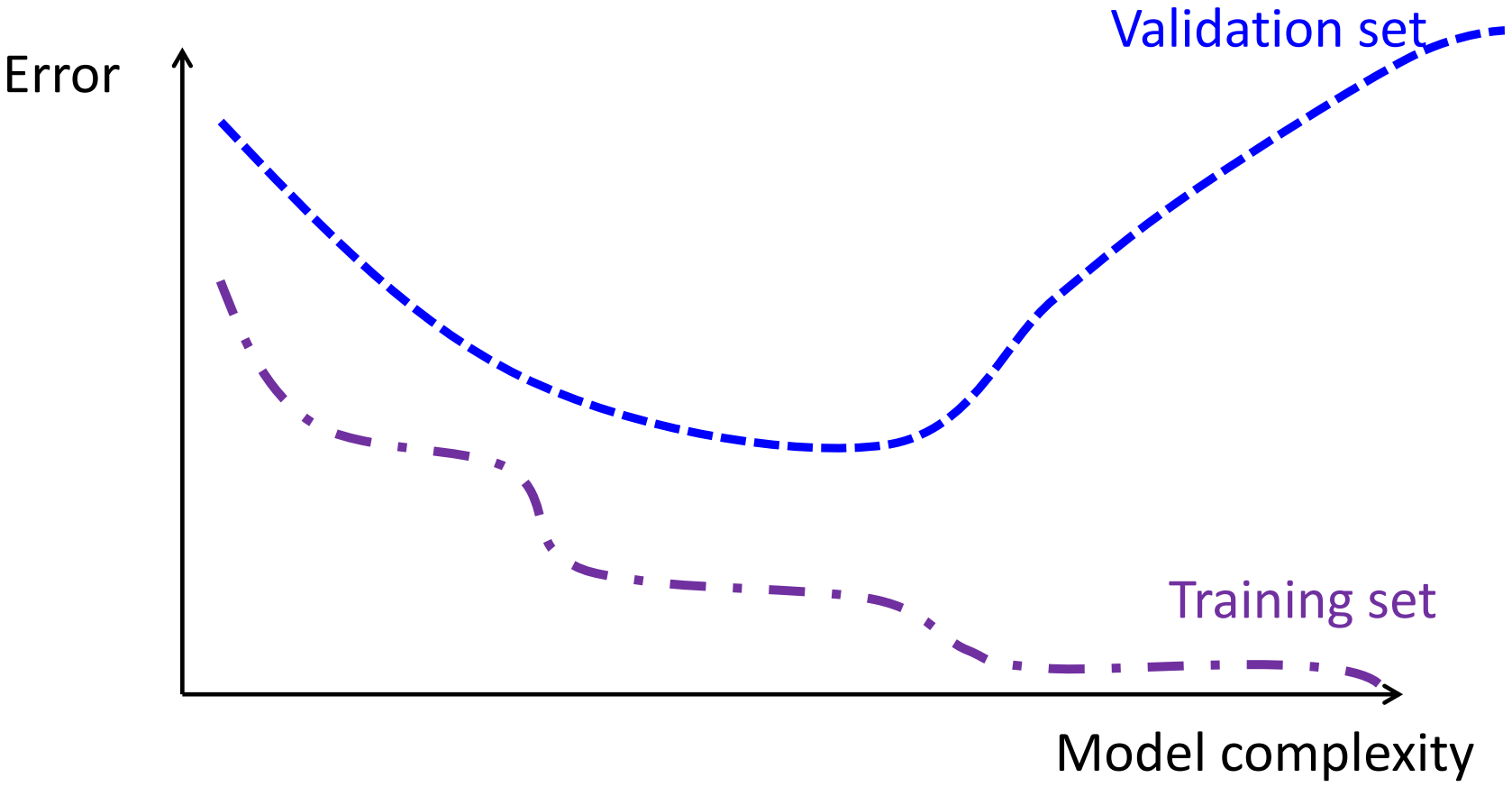
Recall Underfitting vs. Overfitting



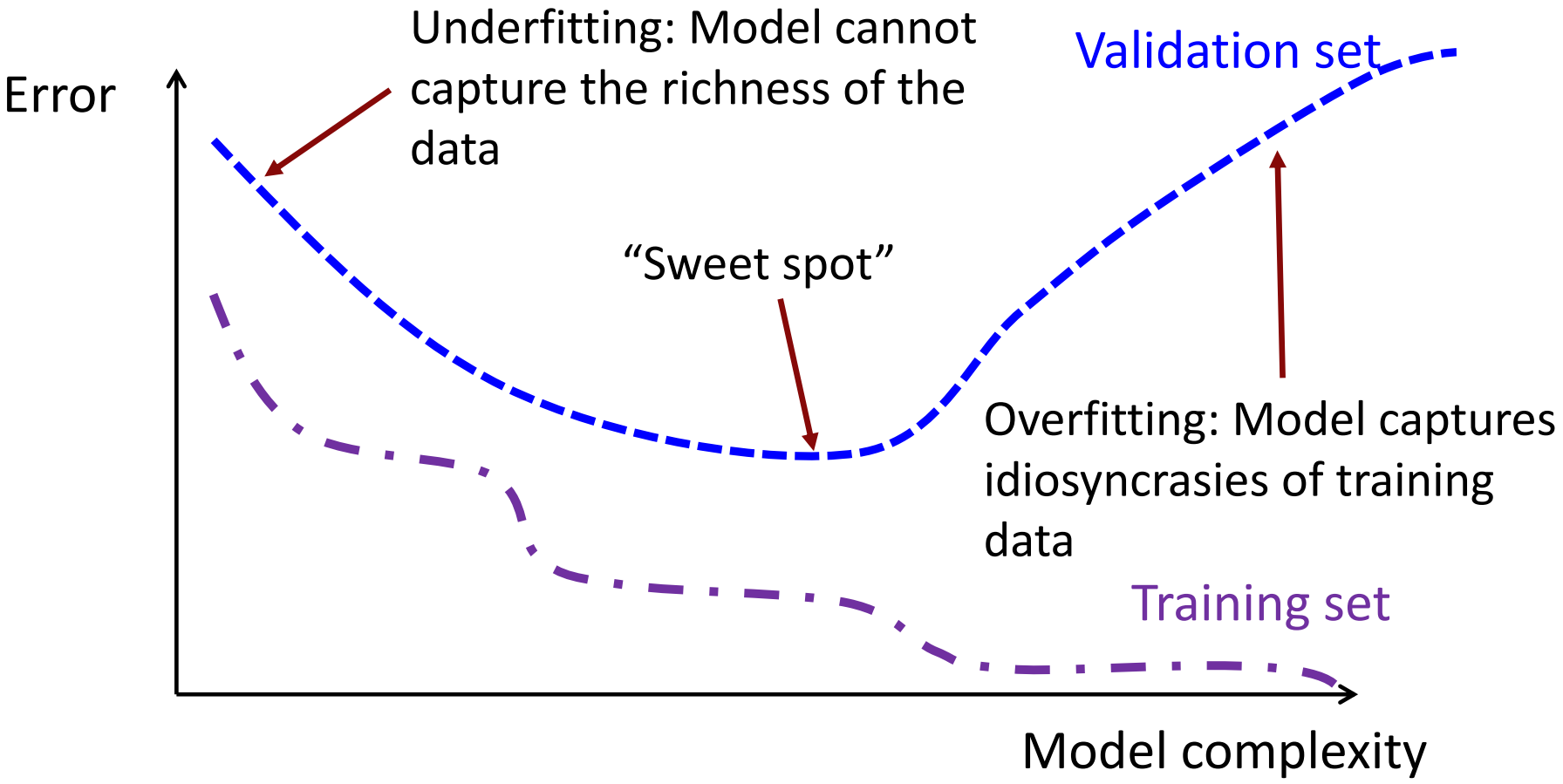
Recall Underfitting vs. Overfitting



Recall Underfitting vs. Overfitting



Recall Underfitting vs. Overfitting

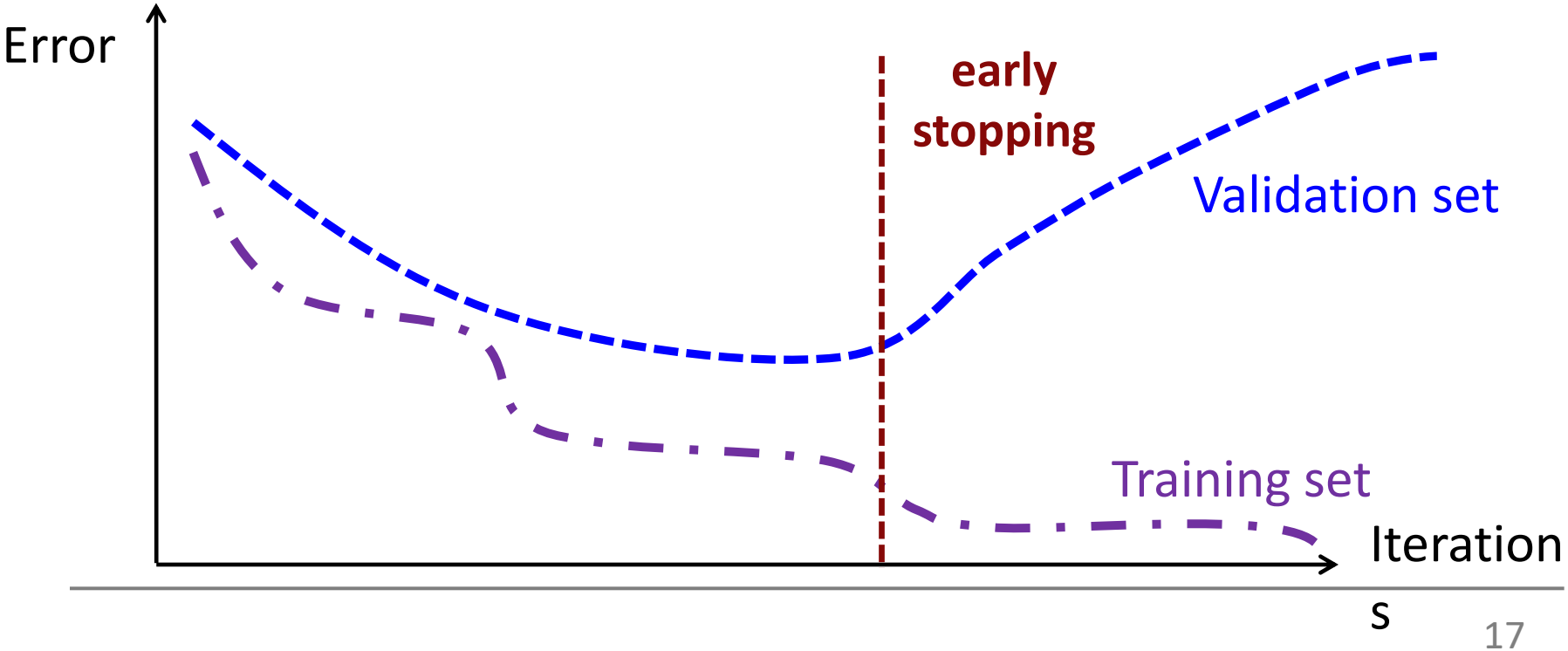


Overfitting in Neural Networks

- To learn smart representations of complex, unstructured data, the NN needs to have large “capacity” i.e., many layers and many neurons in each layer
- But this raises the likelihood of overfitting so we need to add *regularization*
- Several regularization methods have been developed to address this problem

Regularization strategy: *Early Stopping*

Stop the training early before the training loss is minimized by monitoring the loss on a validation dataset.



We will cover this in Lecture 4

Regularization strategy: *Dropout*

Randomly zero out the output from some of the nodes (typically 50% of the nodes) in a hidden layer (implemented as a "dropout layer" in Keras)

